

## MILESTONE 1 – COVER PAGE

TeamNumber: 

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| 22 |
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Please list full names and MacID's of all *present* Team Members

| Full Name:         | MacID:   |
|--------------------|----------|
| Ruidi Liu          | liu127   |
| Matthew Ho         | Hom23    |
| Sharanya Srirangam | srirans  |
| Mayar Aljayoush    | Aljayoum |

# MILESTONE 1 (STAGE 1) – FOCUS AND DEMOGRAPHIC

TeamNumber: 

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## Area of Focus

Indicate your team's chosen **area of focus** in the space below. There are two areas of focus you may choose from:

- Personal monitoring devices
- Technology for automating tasks in the home

|                                  |                            |
|----------------------------------|----------------------------|
| <b>Chosen Area<br/>of Focus:</b> | Personal Monitoring Device |
|----------------------------------|----------------------------|

## Customer Demographic

Indicate the **customer demographic** your team will be targeting in the space below.

- Your customer demographic may be *broad* or *specific*, and may be described in terms of a group of individuals (e.g., patients recovering from at home from a surgery) or a segment of the population (e.g., elderly with a particular disability)

|                                  |                                                 |
|----------------------------------|-------------------------------------------------|
| <b>Customer<br/>Demographic:</b> | Elderly patients suffering from muscle atrophy. |
|----------------------------------|-------------------------------------------------|

## MILESTONE 1 (STAGE 2) – STATEMENTS OF FACTS

Team Number: 

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As a team, list your **statements of facts** in the space below. As there is an expectation that you conduct some preliminary research and market analysis, each statement of fact should be supported by evidence. That is, it should be properly referenced (using IEEE standards) and should not be a made up 'fact'.

Could be focused around the chosen customer or alternatively focused on a task/activity

More than 4

- Elderly people commonly suffer from muscle atrophy. 9.5% - 50% of people 65 years of age and older suffer from sarcopenia worldwide [1].
- Patients with muscle atrophy have trouble picking up objects due to poor muscle coordination [2].
- Patients with muscle atrophy tend to have balance problems [2].
- Burns, strokes injuries or having implants can cause muscle atrophy because of the inactivity for a long period [3].
- Patients with muscle atrophy experience muscle immobility and weakness [4].
- Poor protein and low calories diets can increase the muscle atrophy rather than improve it even while receiving treatment and doing exercise [5].
- Sarcopenia in older adults is a prevalent risk factor for falls and fractures [6].
- Patients with muscle atrophy often feel pain and fatigue [7].
- Muscle atrophy leads to reduced and limited physical fitness [8].

List any references from the statements you came up with as a team in the space below.

- [1] S.F. Chang, "Sarcopenia in the elderly: diagnosis and treatment," *Hu Li Za Zhi*, 2014 Apr; 61(2):101-5. Chinese. doi: 10.6224/JN.61.2.101. PMID: 24676957.
- [2] "Muscle Atrophy", *Healthgrades*, 2021. [Online]. Available: <https://www.healthgrades.com/right-care/bones-joints-and-muscles/muscle-atrophy>. [Accessed: 20- Jan- 2021].
- [3] A. Pietrangelo, "What Causes Muscle Wasting?", *Healthline*, Aug. 2019. [Online]. Available: <https://www.healthline.com/health/muscle-atrophy>. [Accessed: 20- Jan- 2021].
- [4] S. K. Powers, G. S. Lynch, K. T. Murphy, M. B. Reid, and I. Zijdwind, "Disease - Induced Skeletal Muscle Atrophy and Fatigue," *Medicine and science in sports and exercise*, vol. 48, no. 11, pp. 2307–2319, Nov. 2016, doi: 10.1249/MSS.0000000000000975.
- [5] J. M. Argilés, N. Campos, J. M. Lopez-Pedrosa, R. Rueda, and L. Rodriguez-Mañas, "Skeletal Muscle Regulates Metabolism via Interorgan Crosstalk: Roles in Health and Disease," *J. Am. Med. Dir. Assoc.*, vol. 17, no. 9, pp. 789–796, 2016, doi: 10.1016/j.jamda.2016.04.019.
- [6] S. S. Y. Yeung *et al.*, "Sarcopenia and its association with falls and fractures in older adults: A systematic review and meta-analysis," *J. Cachexia. Sarcopenia Muscle*, vol. 10, no. 3, pp. 485–500, 2019, doi: 10.1002/jcsm.12411.
- [7] Q. Liu, J. Gao, J. Deng, and J. Xiao, "Current Studies and Future Directions of Exercise Therapy for Muscle Atrophy Induced by Heart Failure," *Front. Cardiovasc. Med.*, vol. 7, no. October, pp. 1–6, 2020, doi: 10.3389/fcvm.2020.593429.
- [8] E. Kouidi, M. Albani, K. Natsis, et al. "The effects of exercise training on muscle atrophy in haemodialysis patients." *Nephrology, dialysis, transplantation : official publication of the European Dialysis and Transplant Association - European Renal Association*, vol. 13,3 (1998): 685-99. doi:10.1093/ndt/13.3.685

## MILESTONE 1 (STAGE 3) – LIST OF CRITERIA

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As a team, develop a list of criteria that your device is required to meet. List your criteria in the space below. These criteria are akin to *customer requirements*. As such, you should indicate whether each criteria is an objective, function, or constraint. As a reminder, requirements **can be more than one** of the three.

More than 4, non-tech, solution neutral

List your criteria in this field. Bullet-point format is acceptable. For each criteria, indicate (in parentheses) whether it is an objective, function, or constraint.  
**Objective: Attribute or behaviour that the design should have**

***Constraint: Limit or restriction on the design's behaviours or attribute***

***Functions: Solution-neutral engineering actions that the product will perform***

- To reduce muscle atrophy (function)
- To improve quality of life (function)
- Be secure (objective)
- Be reliable (objective)
- Non-irritating (constraint)
- Hypoallergenic (constraint)
- Easy to put on/take off (objective)
- Communicate data to the user/caretaker (function)
- Small size (constraint)
- Lightweight (constraint)

## MILESTONE 1 (STAGE 4) – NEED STATEMENT

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Write your Need Statement in the space below. Recall that your need statement should:

- Have a clearly defined problem (what is the need?)
- Indicate your end-user (who has the need?)
- Have a clearly defined outcome (what do you hope to solve and why is it important?)

|                            |                                                                                                                                                                                                                       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NEED<br/>STATEMENT:</b> | Design a wearable personal monitoring device for elderly patients that are suffering from muscle atrophy. The outcome of the device will help improve their muscle activity, thereby enhancing their quality of life. |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|